Conditional Probability

COMP2711 Discrete Mathematical Tools

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Review: Outcomes, events, and probability

Outcome ω — One possible result of the experiment.

Sample space S — All possible outcomes.

Event $E \subseteq S$ — The event happens when $\omega \in E$.

General discrete model: add the probabilities of outcomes in E.

$$\Pr[E] = \sum_{\omega \in E} \Pr[\{\omega\}].$$

Classical model: if S is finite, nonempty, and outcomes are equally likely,

$$\Pr[E] = \frac{|E|}{|S|}.$$

Review: Event notation

Events are sets of outcomes.

Words	Set notation	Meaning
E and F	$E \cap F$	Both events happen.
E or F	$E \cup F$	At least one happens.
Not E	$\overline{E} = S \setminus E$	E does not happen.

Throughout these slides, we write **and** and **or** for $\cap$ and $\cup$.

$$\Pr[E \text{ and } F] = \Pr[E \cap F].$$

Review: Probability axioms and rules

Probability axioms

1. Nonnegative: $\Pr[E] \geq 0$.
2. The whole sample space: $\Pr[S] = 1$.
3. For disjoint events $E_1, E_2, \ldots$:

$$\Pr[E_1 \text{ or } E_2 \text{ or } \cdots] = \sum_i \Pr[E_i].$$

Useful consequences

$$\Pr[\overline{E}] = 1 - \Pr[E].$$

$$\Pr[E \text{ or } F] = \Pr[E] + \Pr[F] - \Pr[E \text{ and } F].$$

Two children, two conditions

A mother has two children.

Suppose we know the older child is a boy.

What is the chance that the younger child is a boy?

Suppose we know at least one child is a boy.

What is the chance that the younger child is a boy?

Model: independent children; boy/girl equally likely.

Two children, two conditions

A mother has two children.

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BB *BG* *GB* *GG*

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One of two possibilities: $\frac{1}{2}$.

Suppose we know at least one child is a boy.

What is the chance that the younger child is a boy?

BB | *BG* | *GB* | *GG*

Two of three possibilities: $\frac{2}{3}$.

Model: independent children; boy/girl equally likely.

Two children, two conditions

A mother has two children.

Suppose we know the older child is a boy.

What is the chance that the younger child is a boy?

BB | *BG* | *GB* | *GG*

One of two possibilities: $\frac{1}{2}$.

Suppose we know at least one child is a boy born on Wednesday.

What is the chance that the other child is a boy?

Model: independent children; boy/girl equally likely.

Independent, uniform birth weekdays; condition on families. Solution in the note.

Two children, two conditions

A mother has two children.

Suppose we know the older child is a boy.

What is the chance that the younger child is a boy?

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One of two possibilities: $\frac{1}{2}$.

Suppose we know at least one child is a boy born on Wednesday.

What is the chance that the other child is a boy?

$\frac{13}{27}$ **What?**

Model: independent children; boy/girl equally likely.

Independent, uniform birth weekdays; condition on families. Solution in the note.

Lecture roadmap

New information can change a probability.

1. Conditional probability	Use the given information to restrict the possibilities.
2. Multiplication rule	Use updated probabilities to combine events.
3. Independence	Check whether the probability stays the same.
4. Total probability	Split into cases; multiply within each, then add.

SECTION 1

Conditional probability

Combining events

Let E and F be events.

E and F

Both E and F happen.

E or F

At least one of E, F happens, possibly both.

Conditional probability

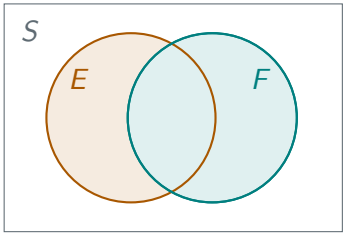
MUST UNDERSTAND

$$\Pr[E \mid F]$$

The probability that E occurs, **given that F occurred**.

$$\Pr[E \mid F] = \frac{\Pr[E \text{ and } F]}{\Pr[F]}, \qquad \Pr[F] > 0.$$

Conditional probability: A visual explanation

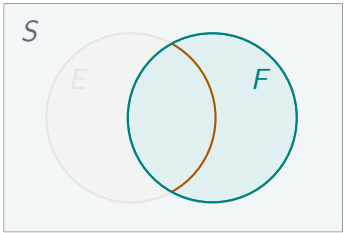


Start with all possible outcomes.

E is the event we are interested in.

Now suppose we learn that F happened.

Conditional probability: A visual explanation

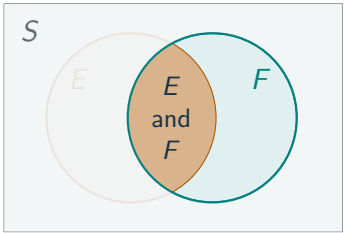


Keep only the outcomes in F.

Outcomes outside F are no longer possible.

Within F, how much probability belongs to E?

Conditional probability: A visual explanation



Keep F, then rescale.

$$\Pr[E \mid F] = \frac{\Pr[E \text{ and } F]}{\Pr[F]}.$$

Divide by $\Pr[F]$ so the remaining possibilities have total probability 1.

Assume $\Pr[F] > 0$.

Classical model: keep the outcomes in F, then count.

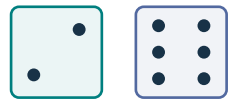
$$\Pr[E \mid F] = \frac{\text{number of outcomes in both } E \text{ and } F}{\text{number of outcomes in } F}.$$

Example: Two dice

Roll two fair dice independently.

Each number from 1 to 6 is **equally likely**.

Each ordered pair has probability 1/36.



$$\Pr[\text{the sum is even} \mid \text{at least one die shows 6}] = ?$$

$$\Pr[\text{at least one die shows 6} \mid \text{the sum is even}] = ?$$

Two dice: Given at least one 6

F: at least one 6.

$$\Pr[E \mid F] = \frac{\Pr[E \text{ and } F]}{\Pr[F]}$$

$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$(1,5)$	$(1,6)$
$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(2,5)$	$(2,6)$
$(3,1)$	$(3,2)$	$(3,3)$	$(3,4)$	$(3,5)$	$(3,6)$
$(4,1)$	$(4,2)$	$(4,3)$	$(4,4)$	$(4,5)$	$(4,6)$
$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$	$(5,5)$	$(5,6)$
$(6,1)$	$(6,2)$	$(6,3)$	$(6,4)$	$(6,5)$	$(6,6)$

Keep the last row or column.

$$|F| = 6 + 6 - 1 = 11.$$

Count $(6,6)$ only once.

Two dice: Given at least one 6

F: at least one 6.

E: even sum.

$(1,1)$	$(1,2)$	$(1,3)$	$(1,4)$	$(1,5)$	$(1,6)$
$(2,1)$	$(2,2)$	$(2,3)$	$(2,4)$	$(2,5)$	$(2,6)$
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Keep the last row or column.

$$|F| = 6 + 6 - 1 = 11.$$

Count $(6,6)$ only once.

Of these, five have an even sum.

$$|E \cap F| = 5.$$

Two dice: Given at least one 6

F: at least one 6.

E: even sum.

$$\Pr[E \mid F] = \frac{\Pr[E \text{ and } F]}{\Pr[F]}$$

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Keep the last row or column.

$$|F| = 6 + 6 - 1 = 11.$$

Count $(6,6)$ only once.

Of these, five have an even sum.

$$|E \cap F| = 5.$$

$$\Pr[E \mid F] = \frac{5/36}{11/36} = \frac{5}{11}.$$

Two dice: Given an even sum

E: even sum.

F: at least one 6.

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$$\Pr[F \mid E] = \frac{\Pr[F \text{ and } E]}{\Pr[E]}$$

Keep pairs with an even sum.

$$|E| = \underbrace{3 \cdot 3}_{\text{both odd}} + \underbrace{3 \cdot 3}_{\text{both even}} = 18.$$

Two dice: Given an even sum

E: **even sum.**

F: at least one 6.

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Keep pairs with an even sum.

$$|E| = \underbrace{3 \cdot 3}_{\text{both odd}} + \underbrace{3 \cdot 3}_{\text{both even}} = 18.$$

Of these, five contain a 6.

$$|F \cap E| = 5.$$

Two dice: Given an even sum

E: **even sum.**

F: at least one 6.

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Keep pairs with an even sum.

$$|E| = \underbrace{3 \cdot 3}_{\text{both odd}} + \underbrace{3 \cdot 3}_{\text{both even}} = 18.$$

Of these, five contain a 6.

$$|F \cap E| = 5.$$

$$\Pr[F \mid E] = \frac{5/36}{18/36} = \frac{5}{18}.$$

Interpretation: Not necessarily causation

Causation means making something happen.

A causal direction

$$\Pr[\text{alarm sounds} \mid \text{fire}]$$

Given there is a fire, how likely is the alarm to sound?

A fire can trigger the alarm.

Interpretation: Not necessarily causation

Causation means making something happen.

A causal direction

$$\Pr[\text{alarm sounds} \mid \text{fire}]$$

Given there is a fire, how likely is the alarm to sound?

A fire can trigger the alarm.

Not a causal direction

$$\Pr[\text{fire} \mid \text{alarm sounds}]$$

Given the alarm sounds, how likely is there to be a fire?

An alarm does not cause a fire.

Given does not mean **caused by**.

Interpretation: No required time order

Our observation: **it rains today**.

Prediction

$$\Pr[\text{rain tomorrow} \mid \text{rain today}]$$

Given it rains today, how likely is rain tomorrow?

Interpretation: No required time order

Our observation: **it rains today**.

Prediction

$$\Pr[\text{rain tomorrow} \mid \text{rain today}]$$

Given it rains today, how likely is rain tomorrow?

Inference about the past

$$\Pr[\text{rain yesterday} \mid \text{rain today}]$$

Given it rains today, how likely is it that it rained yesterday?

The same observation can inform us about the **future** or the **past**.

Example: A five-card hand

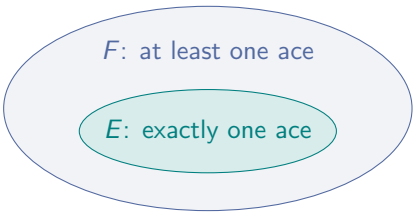
Draw five cards uniformly from a standard 52-card deck, without replacement. Order does not matter.

Given **exactly one ace**, what is the probability of **at least one ace**?

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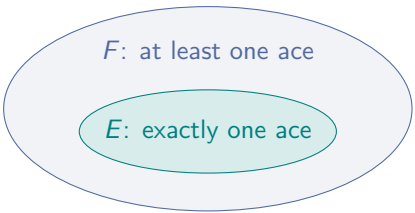
Every hand in E is also in F: $E \subseteq F$.

$$\Pr[F \mid E] = \frac{|F \text{ and } E|}{|E|} = \frac{|E|}{|E|} = 1.$$

Five-card hand: Reverse the condition

A uniformly chosen five-card hand from a standard 52-card deck.

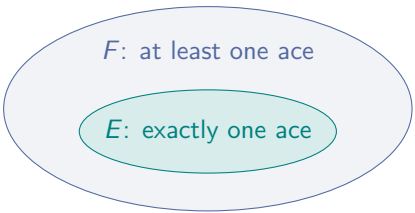
Given **at least one ace**, what is the probability of **exactly one ace**?



Five-card hand: Reverse the condition

A uniformly chosen five-card hand from a standard 52-card deck.

Given **at least one ace**, what is the probability of **exactly one ace**?



Keep all hands in F. Only those in E qualify.

$$\Pr[E \mid F] = \frac{|E \text{ and } F|}{|F|} = \frac{|E|}{|F|}.$$

Five-card hand: Count within the condition

F: at least one ace. E: exactly one ace.

hands with at least one ace = # all hands − # hands with no aces.

$$|F| = \underbrace{\binom{52}{5}}_{\text{all hands}} - \underbrace{\binom{48}{5}}_{\text{choose 5 from 48 non-ace cards}}.$$

Five-card hand: Count within the condition

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$$|F| = \underbrace{\binom{52}{5}}_{\text{all hands}} - \underbrace{\binom{48}{5}}_{\text{choose 5 from 48 non-ace cards}}.$$

Desired hands: choose one of the four aces and four of the 48 non-aces.

$$|E \text{ and } F| = |E| = \binom{4}{1}\binom{48}{4}.$$

Five-card hand: Count within the condition

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Desired hands: choose one of the four aces and four of the 48 non-aces.

$$|E \text{ and } F| = |E| = \binom{4}{1}\binom{48}{4}.$$

$$\Pr[E \mid F] = \frac{|E \text{ and } F|}{|F|} = \frac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5} - \binom{48}{5}}.$$

Follow-up: Aces and kings

Draw a five-card hand uniformly from a standard 52-card deck.

$$\Pr[\text{at least one ace} \mid \text{at least one king}] = ?$$

Hint: Use the inclusion–exclusion principle.

SECTION 2

The multiplication rule

Multiplication rule

MUST UNDERSTAND

For $\Pr[E] > 0$, start from

$$\Pr[F \mid E] = \frac{\Pr[E \text{ and } F]}{\Pr[E]}.$$

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For $\Pr[E] > 0$, start from

$$\Pr[F \mid E] = \frac{\Pr[E \text{ and } F]}{\Pr[E]}.$$

$$\boxed{\Pr[E \text{ and } F] = \Pr[E] \Pr[F \mid E]}.$$

Multiplication rule

MUST UNDERSTAND

For $\Pr[E] > 0$, start from

$$\Pr[F \mid E] = \frac{\Pr[E \text{ and } F]}{\Pr[E]}.$$

$$\boxed{\Pr[E \text{ and } F] = \Pr[E] \Pr[F \mid E]}.$$

We may reverse the order, if $\Pr[F] > 0$:

$$\Pr[E \text{ and } F] = \Pr[F] \Pr[E \mid F].$$

Example: Two aces

Draw two cards uniformly **without replacement**.

What is the probability that both are aces?

$$\Pr[\text{First Ace}] = \frac{4 \text{ aces}}{52 \text{ cards}} = \frac{1}{13}.$$

Example: Two aces

Draw two cards uniformly **without replacement**.

What is the probability that both are aces?

$$\Pr[\text{First Ace}] = \frac{4 \text{ aces}}{52 \text{ cards}} = \frac{1}{13}.$$

$$\Pr[\text{Second Ace} \mid \text{First Ace}] = \frac{3 \text{ aces}}{51 \text{ cards}} = \frac{1}{17}.$$

Example: Two aces

Draw two cards uniformly **without replacement**.

What is the probability that both are aces?

$$\Pr[\text{First Ace}] = \frac{4 \text{ aces}}{52 \text{ cards}} = \frac{1}{13}.$$

$$\Pr[\text{Second Ace} \mid \text{First Ace}] = \frac{3 \text{ aces}}{51 \text{ cards}} = \frac{1}{17}.$$

$$\Pr[\text{Both aces}] = \Pr[\text{First Ace}] \times \Pr[\text{Both aces} \mid \text{First Ace}]$$

Example: Two aces

Draw two cards uniformly **without replacement**.

What is the probability that both are aces?

$$\Pr[\text{First Ace}] = \frac{4 \text{ aces}}{52 \text{ cards}} = \frac{1}{13}.$$

$$\Pr[\text{Second Ace} \mid \text{First Ace}] = \frac{3 \text{ aces}}{51 \text{ cards}} = \frac{1}{17}.$$

$$\begin{aligned}\Pr[\text{Both aces}] &= \Pr[\text{First Ace}] \times \Pr[\text{Both aces} \mid \text{First Ace}] \\ &= \Pr[\text{First Ace}] \times \Pr[\text{Second Ace} \mid \text{First Ace}]\end{aligned}$$

Example: Two aces

Draw two cards uniformly **without replacement**.

What is the probability that both are aces?

$$\Pr[\text{First Ace}] = \frac{4 \text{ aces}}{52 \text{ cards}} = \frac{1}{13}.$$

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$$\begin{aligned}\Pr[\text{Both aces}] &= \Pr[\text{First Ace}] \times \Pr[\text{Both aces} \mid \text{First Ace}] \\ &= \Pr[\text{First Ace}] \times \Pr[\text{Second Ace} \mid \text{First Ace}] \\ &= \frac{1}{13} \times \frac{1}{17} = \boxed{\frac{1}{221}}.\end{aligned}$$

Two aces: Direct counting

Draw two cards uniformly without replacement. Ignore their order.

All two-card hands

$$\binom{52}{2} = 1326.$$

Two aces from the four aces

$$\binom{4}{2} = 6.$$

Two aces: Direct counting

Draw two cards uniformly without replacement. Ignore their order.

All two-card hands

$$\binom{52}{2} = 1326.$$

Two aces from the four aces

$$\binom{4}{2} = 6.$$

Both methods give the same probability:

$$\Pr[\text{both aces}] = \underbrace{\frac{\binom{4}{2}}{\binom{52}{2}}}_{\text{direct counting}} = \frac{6}{1326} = \frac{1}{221} = \underbrace{\frac{4}{52} \cdot \frac{3}{51}}_{\text{multiplication rule}}.$$

Example: Three aces

Draw three cards uniformly without replacement.

What is the probability that all three are aces?

First ace	Second ace, given first	Third ace, given first two
$4/52$	$3/51$	

Example: Three aces

Draw three cards uniformly without replacement.

What is the probability that all three are aces?

First ace	Second ace, given first	Third ace, given first two
$4/52$	$3/51$	$2/50$

$$\Pr[A_1 \text{ and } A_2 \text{ and } A_3] = \frac{4}{52}\frac{3}{51}\frac{2}{50} = \frac{1}{5525}.$$

A_i: draw i is an ace.

General chain rule

To find the probability that **all n events happen**:

$$\Pr[E_1 \text{ and } E_2 \text{ and } \cdots \text{ and } E_n]$$

$$= \Pr[E_1] \Pr[E_2 \mid E_1] \cdots \Pr[E_n \mid E_1 \text{ and } \cdots \text{ and } E_{n-1}].$$

Each factor gives the probability of the next event, **given that all preceding events happen**.

Each conditioning event must have positive probability:

$$\Pr[E_1 \text{ and } \cdots \text{ and } E_{i-1}] > 0, \qquad i = 2, \ldots, n.$$

SECTION 3

Independence and complements

Independent events

MUST UNDERSTAND

Definition: Events E and F are independent when

$$\boxed{\Pr[E \text{ and } F] = \Pr[E] \cdot \Pr[F]}.$$

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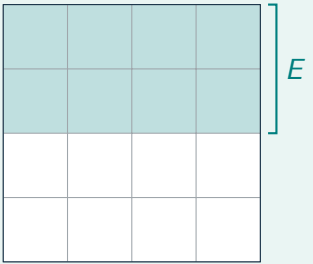
Interpretation: Learning F does not change the probability of E.

For $\Pr[F] > 0$,

$$\Pr[E \mid F] = \frac{\Pr[E \text{ and } F]}{\Pr[F]} = \frac{\Pr[E] \cdot \Pr[F]}{\Pr[F]} = \Pr[E].$$

Independent events: A visual explanation

Each square is one equally likely outcome.

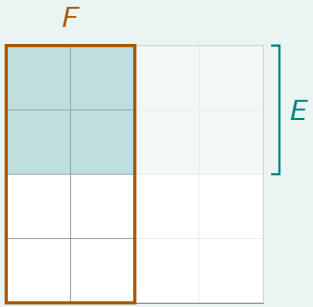


Before learning F

$$\Pr[E] = \frac{8}{16} = \frac{1}{2}.$$

Independent events: A visual explanation

Each square is one equally likely outcome.



Before learning F

$$\Pr[E] = \frac{8}{16} = \frac{1}{2}.$$

Given F: keep the left two columns

$$\Pr[E \mid F] = \frac{4}{8} = \frac{1}{2}.$$

The same proportion belongs to E: **its probability has not changed**.

Example: An ace and a heart

Draw one card uniformly. Let A = ace and H = heart.

Does knowing the card is a heart change its chance of being an ace?

	♡	♢	♣	♠
Aces	1	1	1	1
Other cards	12	12	12	12

Example: An ace and a heart

Draw one card uniformly. Let A = ace and H = heart.

Does knowing the card is a heart change its chance of being an ace?

	♡	♢	♣	♠
Aces	1	1	1	1
Other cards	12	12	12	12

$$\Pr[A \mid H] = \frac{1}{13} = \frac{4}{52} = \Pr[A].$$

Independent: $\Pr[A \text{ and } H] = \frac{1}{52} = \frac{4}{52}\frac{13}{52}$.

Replacement and independence

Draw two cards uniformly. Let A_i = draw i is an ace.

With replacement: return the first card and reshuffle.

Experiment	$\Pr[A_2]$	$\Pr[A_2 \mid A_1]$	Independent?
With replacement	$4/52$	$4/52$	
Without replacement	$4/52$	$3/51$	

Replacement and independence

Draw two cards uniformly. Let A_i = draw i is an ace.

With replacement: return the first card and reshuffle.

Experiment	$\Pr[A_2]$	$\Pr[A_2 \mid A_1]$	Independent?
With replacement	$4/52$	$4/52$	**Yes**
Without replacement	$4/52$	$3/51$	No

Before any card is seen, each card is equally likely to occupy the second position.

$$\Pr[A_2] = \frac{4}{52}, \qquad \Pr[A_2 \mid A_1] = \frac{3}{51} \neq \frac{4}{52} \quad \text{without replacement.}$$

Disjoint is not independent

Draw one card uniformly.

Let H = heart and S = spade.

Are these events independent?



Disjoint is not independent

Draw one card uniformly.

Let H = heart and S = spade.

Are these events independent?



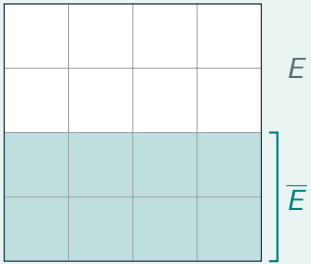
$$\Pr[H \text{ and } S] = 0, \qquad \Pr[H \mid S] = 0 \neq \frac{1}{4} = \Pr[H].$$

Disjoint events with positive probabilities are dependent.

Independence and complements

If E, F are independent, then $\overline{E}, F$ are independent.

$\overline{E}$ means E does not occur.



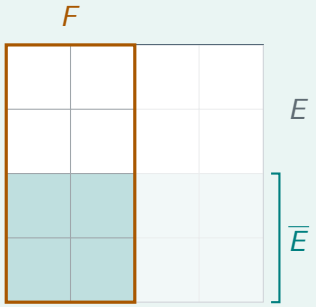
Before learning F

$$\Pr[\overline{E}] = \frac{8}{16} = \frac{1}{2}.$$

Independence and complements

If E, F are independent, then $\overline{E}, F$ are independent.

$\overline{E}$ means E does not occur.



Before learning F

$$\Pr[\overline{E}] = \frac{8}{16} = \frac{1}{2}.$$

Given F: keep the left two columns

$$\Pr[\overline{E} \mid F] = \frac{4}{8} = \frac{1}{2}.$$

If the proportion in E stays the same, **so does the proportion outside E.**

Independence and complements: Proof

If E, F are independent, then $\overline{E}, F$ are independent.

$$\Pr[\overline{E} \text{ and } F] = \Pr[F] - \Pr[E \text{ and } F]$$

Independence and complements: Proof

If E, F are independent, then $\overline{E}, F$ are independent.

$$\begin{aligned}\Pr[\overline{E} \text{ and } F] &= \Pr[F] - \Pr[E \text{ and } F] \\ &= \Pr[F] - \Pr[E]\Pr[F]\end{aligned}$$

Independence and complements: Proof

If E, F are independent, then $\overline{E}, F$ are independent.

$$\begin{aligned}\Pr[\overline{E} \text{ and } F] &= \Pr[F] - \Pr[E \text{ and } F] \\ &= \Pr[F] - \Pr[E]\Pr[F] \\ &= (1 - \Pr[E])\Pr[F] = \Pr[\overline{E}]\Pr[F].\end{aligned}$$

Complementing either or both events preserves independence.

Pairwise and mutual independence

For three events A, B, C:

Pairwise independence

Every pair satisfies the product rule.

$$\Pr[A \text{ and } B] = \Pr[A]\Pr[B],$$
$$\Pr[A \text{ and } C] = \Pr[A]\Pr[C],$$
$$\Pr[B \text{ and } C] = \Pr[B]\Pr[C].$$

Mutual independence

All three pair equalities, **and also**

$$\Pr[A \text{ and } B \text{ and } C]$$
$$= \Pr[A]\Pr[B]\Pr[C].$$

Pairwise and mutual independence

For three events A, B, C:

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Every pair satisfies the product rule.

$$\Pr[A \text{ and } B] = \Pr[A]\Pr[B],$$
$$\Pr[A \text{ and } C] = \Pr[A]\Pr[C],$$
$$\Pr[B \text{ and } C] = \Pr[B]\Pr[C].$$

Mutual independence

All three pair equalities, **and also**

$$\Pr[A \text{ and } B \text{ and } C] = \Pr[A]\Pr[B]\Pr[C].$$

For more events: every selection of two or more events must satisfy the product rule.

Example: Three events from two flips

Flip a fair coin twice independently.

A: first flip is heads.

B: second flip is heads.

C: exactly one flip is heads.

Outcome	A	B	C
HH	Yes	Yes	No
HT	Yes	No	Yes
TH	No	Yes	Yes
TT	No	No	No

$$\Pr[A] = \Pr[B] = \Pr[C] = \frac{1}{2}.$$

Pairwise, but not mutual

Two fair independent flips: A = first H, B = second H, C = exactly one H.

Event	Outcomes	Probability
A and B	HH	$1/4 = (1/2)(1/2)$
A and C	HT	$1/4 = (1/2)(1/2)$
B and C	TH	$1/4 = (1/2)(1/2)$

Every pair is independent.

Pairwise, but not mutual

Two fair independent flips: A = first H, B = second H, C = exactly one H.

Event	Outcomes	Probability
A and B	HH	$1/4 = (1/2)(1/2)$
A and C	HT	$1/4 = (1/2)(1/2)$
B and C	TH	$1/4 = (1/2)(1/2)$

Every pair is independent.

$$\Pr[A \text{ and } B \text{ and } C] = 0 \neq \frac{1}{8} = \Pr[A]\Pr[B]\Pr[C].$$

The three events are not mutually independent.

SECTION 4

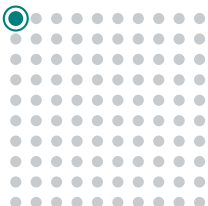
Total probability

Example: A medical test

A person is selected randomly from a population and tested.

Assume the following hypothetical rates.

The population



1 in 100 has the disease.

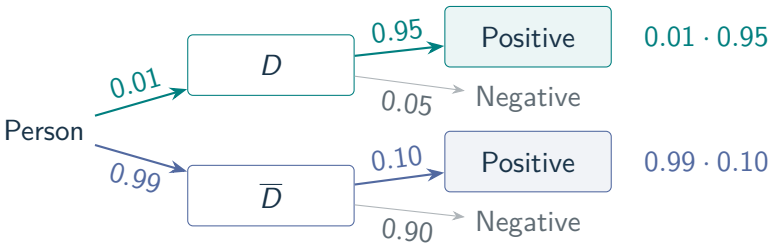
The test

	Tests positive
Has the disease	95%
Does not have the disease	10%

What is the probability of a positive test?

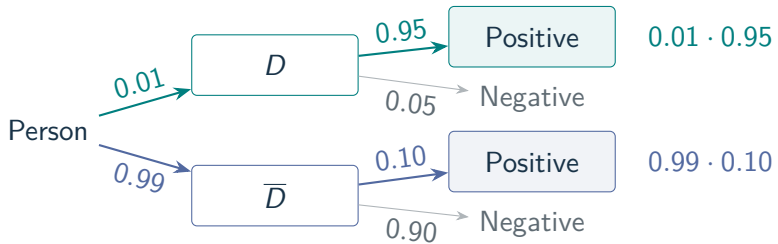
Medical test: Two paths to a positive result

D: has the disease. T: tests positive.



Medical test: Two paths to a positive result

D: has the disease. T: tests positive.

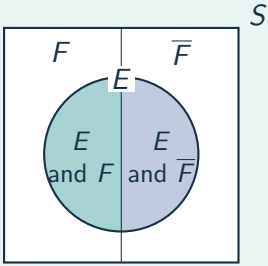


These two cases cannot happen together. Add their probabilities:

$$\Pr[T] = \underbrace{0.01 \cdot 0.95}_{\text{disease and positive}} + \underbrace{0.99 \cdot 0.10}_{\text{no disease and positive}} = \boxed{0.1085 = 10.85\%}.$$

Total probability

MUST UNDERSTAND



Split E into **two disjoint pieces**:

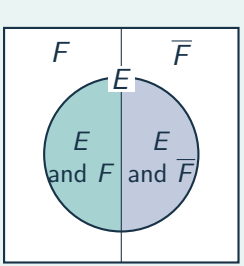
E and F

E and $\overline{F}$

Together, they cover all of E.

Total probability

MUST UNDERSTAND



Split E into **two disjoint pieces**:

E and F

E and $\overline{F}$

Together, they cover all of E.

$$\Pr[E] = \Pr[F]\Pr[E \mid F] + \Pr[\overline{F}]\Pr[E \mid \overline{F}].$$

Assume $\Pr[F] > 0$ and $\Pr[\overline{F}] > 0$.

Total probability: General form

Split the sample space S into **disjoint cases** $F_1, \ldots, F_m$.

These cases cover S, and $\Pr[F_i] > 0$:

$$\boxed{\Pr[E] = \sum_{i=1}^{m} \Pr[F_i] \Pr[E \mid F_i]}.$$

Count the part of E in each case, then add.

Example: Three bags

Choose a bag with the probabilities shown below. Then draw one ball uniformly from that bag.



What is the probability of drawing a red ball?

Three bags: Total probability

R: the ball is red. F_A, F_B, F_C: the chosen bag.

The three bag choices are disjoint and cover all possibilities.

Bag	Chosen	Red, given this bag	Chosen and red
A	0.5	1/4	
B	0.3	2/4	
C	0.2	3/4	

Three bags: Total probability

R: the ball is red. F_A, F_B, F_C: the chosen bag.

The three bag choices are disjoint and cover all possibilities.

Bag	Chosen	Red, given this bag	Chosen and red
A	0.5	$1/4$	$0.5 \cdot 1/4$
B	0.3	$2/4$	$0.3 \cdot 2/4$
C	0.2	$3/4$	$0.2 \cdot 3/4$

Three bags: Total probability

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The three bag choices are disjoint and cover all possibilities.

Bag	Chosen	Red, given this bag	Chosen and red
A	0.5	$1/4$	$0.5 \cdot 1/4$
B	0.3	$2/4$	$0.3 \cdot 2/4$
C	0.2	$3/4$	$0.2 \cdot 3/4$

$$\Pr[R] = \Pr[R \mid F_A] \cdot \Pr[F_A] + \Pr[R \mid F_B] \cdot \Pr[F_B] + \Pr[R \mid F_C] \cdot \Pr[F_C]$$

$$= \frac{1}{4} \cdot 0.5 + \frac{2}{4} \cdot 0.3 + \frac{3}{4} \cdot 0.2 = \boxed{42.5\%}.$$

Summary

Conditional probability $\Pr[E \mid F] = \dfrac{\Pr[E \text{ and } F]}{\Pr[F]}, \quad \Pr[F] > 0.$

Multiplication rule $\Pr[E \text{ and } F] = \Pr[E]\Pr[F \mid E], \quad \Pr[E] > 0.$

Independence Events E and F are independent

$\iff \Pr[E \text{ and } F] = \Pr[E] \cdot \Pr[F].$

Total probability $\Pr[E] = \sum_i \Pr[F_i]\Pr[E \mid F_i].$

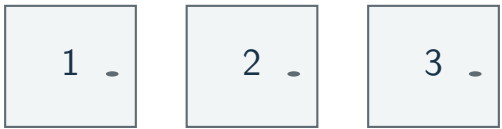
Disjoint cases F_i cover S; each $\Pr[F_i] > 0$.

Optional: Monty Hall

You are on a game show. Three doors stand in front of you.

One hides a prize. The other two are empty.

Each door is equally likely to hide the prize.



Which door would you choose: 1, 2, or 3?

Optional: Monty Hall

You are on a game show. Three doors stand in front of you.

One hides a prize. The other two are empty.

Each door is equally likely to hide the prize.



Suppose you choose **door 1**. The host says:

"Before you open it, let me show you something."

Monty Hall: Stay or switch?

You chose door 1. The host opens **door 3**: it is empty.



Monty Hall: Stay or switch?

You chose door 1. The host opens **door 3**: it is empty.



The host asks:

"Keep door 1, or switch to door 2?"

Monty Hall: Stay or switch?

You chose door 1. The host opens **door 3**: it is empty.



The host asks:

"Keep door 1, or switch to door 2?"

Before answering: how did the host choose which door to open?

Monty Hall: Two host procedures

Same observation: you chose door 1; the host opened an empty door 3.

Case 1: The host knows

He always opens an unchosen empty door.

If both are empty, he chooses either with probability $1/2$.

Case 2: The host chooses randomly

He opens door 2 or 3 with probability $1/2$, independently of the prize.

He might reveal the prize. This time, he did not.

In both cases, after an empty door is opened, he offers a switch.

Monty Hall: Two host procedures

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He opens door 2 or 3 with probability $1/2$, independently of the prize.

He might reveal the prize. This time, he did not.

In both cases, after an empty door is opened, he offers a switch.

Do these procedures give the same information?

Case 1: The host knows

F: the host opens door 3 and it is empty. You chose door 1.

The host avoids the prize; if both unchosen doors are empty, he chooses uniformly.

Prize location	Initial chance	Chance of F given location	Location and F
Door 1	1/3	1/2	1/6
Door 2	1/3	1	1/3
Door 3	1/3	0	0

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Prize location	Initial chance	Chance of F given location	Location and F
Door 1	$1/3$	$1/2$	$1/6$
Door 2	$1/3$	1	$1/3$
Door 3	$1/3$	0	0

$$\Pr[\text{prize at door 2} \mid F] = \frac{1/3}{1/6 + 1/3} = \boxed{\frac{2}{3}}.$$

Switch: door 2 has probability $2/3$; door 1 has probability $1/3$.

Case 2: The host chooses randomly

F: the host opens door 3 and it is empty. You chose door 1.

The host chooses door 2 or 3 uniformly, independently of the prize.

Prize location	Initial chance	Chance of F given location	Location and F
Door 1	1/3	1/2	1/6
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Door 1	$1/3$	$1/2$	$1/6$
Door 2	$1/3$	$1/2$	$1/6$
Door 3	$1/3$	0	0

$$\Pr[\text{prize at door 2} \mid F] = \frac{1/6}{1/6 + 1/6} = \boxed{\frac{1}{2}}.$$

Either choice: given this empty-door reveal, each closed door has probability $1/2$.

Monty Hall: Hidden conditions matter

You chose door 1. The host opened door 3, and it was empty.

Host's procedure	Stay	Switch
Avoids the prize; fair choice when both are empty	1/3	2/3
Chooses randomly; happens to reveal an empty door	1/2	1/2

The same observation can carry different information.

We must specify how that observation was produced.

Practice (1): Conditioning and multiplication

1. One die

Roll a fair die. Let $E = \{2, 4, 6\}$ and $F = \{4, 5, 6\}$.

Find $\Pr[E \mid F]$ and $\Pr[F \mid E]$. List the remaining outcomes in each case.

2. Reasoning backward

A bag has three red and two blue balls. Draw three uniformly without replacement.

Given that the third ball is red, what is the probability that the first two have different colors?

Complete solutions are in the course note.

Practice (2): Independence and probability bounds

3. Test independence

Roll a fair die. Let

$$E = \{2, 4, 6\}, \qquad F = \{4, 5, 6\}, \qquad G = \{1, 2, 3, 4\}.$$

Are E, F independent? Are E, G independent? Justify your answers.

4. Unknown dependence

You know only $\Pr[E] = 0.6$ and $\Pr[F] = 0.5$.

What are the smallest and largest possible values of $\Pr[E \mid F]$?

What if E, F are independent?

Complete solutions are in the course note.

Practice (3): Multiple events and cases

5. Three flips

Flip a fair coin three times independently.

A: first flip is heads; B: second flip is heads;

C: all three results are the same.

Are A, B, C pairwise independent? Mutually independent?

6. Same bag, two draws

A fair coin chooses A (3 red, 1 blue) or B (1 red, 3 blue). Draw twice uniformly from that same bag, replacing and mixing after the first draw.

Let R_i mean draw i is red. Find $\Pr[R_1 \text{ and } R_2]$ and $\Pr[R_2 \mid R_1]$. Are R_1, R_2 independent?

Complete solutions are in the course note.